TURF MANAGER

Product Requirements Document (PRD) — Technical Architecture & Functional Specification

Version: 1.0.0
Target Release: AY 2026 / Capstone
Target Market: Dhaka, Bangladesh

Stack Baseline: MongoDB • Express.js • React 18 (Vite) • Node.js (MERN) | React Leaflet / OpenStreetMap GIS | Two-Phase Lock Simulation | SSLCOMMERZ & bKash Sandbox Gateway

1. EXECUTIVE SUMMARY

Turf Manager is a specialized, mobile-first web platform engineered to eliminate friction in discovering, reserving, and managing artificial football turfs across Dhaka, Bangladesh. Currently, the local sports booking ecosystem relies on fragmented channels—primarily Facebook page direct messaging and manual cellular phone calls. This operational model leads to frequent double-bookings, unverified advance deposits via peer-to-peer mobile financial services (MFS), and a lack of schedule transparency.

Turf Manager provides a unified, three-tiered system:

1. End Consumers (Players)
Discover verified venues via an interactive geospatial map (React Leaflet / OSM), query real-time slot schedules, place 10-minute pessimistic holds during checkout, and execute verified advance deposits.

2. Turf Operators (Owners)
Digitize physical ground schedules, automate time-slot allocations, configure dynamic peak/off-peak pricing rules, and track cash-out versus online ledger receipts.

3. Platform Administrators
Moderate venue onboarding, enforce quality standards, resolve disputed reviews, and monitor high-level system operations.

The application leverages React 18+ (Vite) on the frontend, Node.js/Express.js for the RESTful API tier, MongoDB Atlas with two-phase commit simulation and `2dsphere` geospatial indexing at the persistence layer, and SSLCOMMERZ / bKash gateway integrations backed by a sandboxed mock runner.

2. PROBLEM STATEMENT

Football (sub-format 5-a-side, 6-a-side, and 7-a-side) on artificial turf pitches is widely popular among youth, university students, and corporate professionals across Dhaka's key hubs (e.g., Banani, Dhanmondi, Uttara, Badda, Mirpur, Bashundhara). However, the market faces clear operational bottlenecks:

• Information Fragmentation:
Venues announce availability on disjointed social media feeds without programmatic synchronization. Players spend hours calling 4 to 6 venues to secure a single evening slot.

• Geospatial Inefficiency:
Due to severe road congestion across Dhaka, proximity and actual transit times are critical. Existing platforms lack localized maps showing precise turf dimensions, parking availability, and locker room quality.

• Payment & Booking Conflicts:
Advance deposits are transacted via unstructured personal bKash/Nagad transfers with manual screenshot verification. Lack of atomic reservation logic leads to race conditions where two distinct groups pay for the same pitch-hour simultaneously.

• Zero Tooling for Turf Owners:
Operators track multi-pitch facilities using physical notebooks or rudimentary spreadsheets. This makes it difficult to adjust peak pricing, manage pitch maintenance, or prevent employee leakage.

3. PRODUCT VISION AND GOALS

3.1 Vision Statement: To establish the primary digital infrastructure for recreational sports facilities across Dhaka, replacing manual booking friction with verified schedules, instant geospatial discovery, and safe digital payments.

3.2 Measurable Project Goals

CORE LATENCY	DATA INTEGRITY	MOBILE EXPERIENCE	MAINTAINABILITY
< 250ms	**0 Overlaps**	**≥ 44×44 px**	**REST Modular**
Search & bounding-box geo queries inside Dhaka networks.	Isolated atomic slot reservations eliminate double-bookings.	Touch targets & sub-2s load times on 4G cellular connections.	Clean separation of concerns suitable for capstone evaluation.

4. TARGET USERS AND PERSONAS

Persona A: Organizer — Customer

Tanvir Ahmed, 21 • Private University Student

Context: Organizes weekly 6-a-side matches with classmates every Thursday or Friday evening on a budget of BDT 1,500 – BDT 2,500 per team-hour.

Pain Points: Spends hours calling turf managers who do not pick up; risks losing money via unverified personal bKash transfers; lacks parking and boot rental info.

Goals: Filter turfs by area (Mirpur/Uttara), check real-time slots for 8:00 PM, hold slot for 10 mins while collecting cash, and book securely.

Persona B: Facility Operator — Turf Owner

Rafiqul Islam, 38 • 2-Pitch Venue Partner (Badda)

Context: Operates daily from 6:00 AM to 2:00 AM. Manages peak evening demand and daytime slumps with two shift managers answering phones.

Pain Points: Staff mistakenly double-book peak 8:00 PM–10:00 PM slots; customers fail to show up after claiming to pay at counter; manual TrxID verification is tedious.

Goals: Digitize both pitches, establish distinct peak floodlight vs. off-peak pricing, review verified deposits, and block slots for pitch maintenance.

Persona C: Platform Admin — Administrator

Fahmida Rahman, 22 • Student Lead Architect

Context: Responsible for verifying venue listings, monitoring platform uptime, and ensuring data integrity across the student project.

Pain Points: Illegitimate venues spamming fake pitches; user disputes over refund eligibility; unauthorized users attempting access to operational endpoints.

Goals: Review submitted registration documents, inspect uploaded venue images, ban abusive accounts, monitor system health, and inspect audit logs.

4.4 Role-Based Access Control (RBAC) Matrix

Feature / Domain Capability	Customer	Turf Owner	Platform Admin
Browse Active Listings & View Dhaka Map	Read	Read	Read
Search & Filter Verified Slots	Read	Read	Read
Hold & Book Available Slot (10-Min Pessimistic Lock)	Execute	Denied	Denied
Submit Review on Completed Booking	Execute	Denied	Moderate
Register New Turf Listing	Denied	Create	Create
Update Owned Turf Profile & Operating Hours	Denied	Update (Owned)	Update (Any)
Block Turf Slots for Maintenance / Manual Walk-in	Denied	Execute (Owned)	Execute (Any)
Review Turf Revenue & Booking Metrics	Denied	Read (Owned)	Read (System-wide)
Approve / Reject Pending Turf Registrations	Denied	Denied	Execute
Suspend / Delete Accounts or Listings	Denied	Denied	Execute

5. SCOPE AND EXCLUSIONS

[IN SCOPE: Core MVP & Academic Baseline]

- Verified registration & role-segregated JWT auth
- Turf discovery with Dhaka-centric React Leaflet map integration
- 10-minute checkout slot-hold engine with concurrency controls
- Sandboxed MFS / SSLCOMMERZ gateway runner for localized payment
- Turf profile & slot configuration (peak vs off-peak rates)
- Post-match ratings and reviews for verified bookings
- Role-specific administrative moderation tools

[EXCLUDED: Out of Scope / Future Work]

- Multi-sport support (cricket, tennis, badminton)
- Cross-city geographic expansion beyond Dhaka metropolitan area
- Native iOS/Android application builds (PWA prioritized)
- Real-time automated split-bill payment engines among teammates
- Live GPS hardware tracking or IoT turnstiles on pitch
- Full automated accounting / statutory tax reconciliation
- Complex tournament bracket engines & automated refund bank payouts

6. SYSTEM ARCHITECTURE AND DOMAIN MODELING

6.1 Architectural Topology: The application uses a 3-tier client-server architecture deployed on cloud infrastructure:

CLIENT TIER (React 18 / Vite)
- Tailwind CSS / Mobile-First UI
- React Leaflet / OSM Tiles
- Axios with JWT Interceptors
- React Hook Form + Zod Validation

APPLICATION TIER (Node/Express)
- Middleware: CORS, Helmet, RateLimiter
- Domains: Auth, Search/Geo, HoldEngine
- Agenda / Cron Slot Lock Reversion
- Webhook HMAC Signature Verifier

PERSISTENCE & CLOUD TIER
- MongoDB Atlas: `2dsphere` & ACID
- Cloudinary CDN: Multi-res pitch photos
- SSLCOMMERZ / bKash Sandbox
- Server-to-Server IPN Webhook

6.2 Entity-Relationship Domain Model (ERD)
The domain encapsulates six core entities linked through strong relational integrity and transactional references:

- **USER (1) ── (N) TURF:** A user with role `TURF_OWNER` registers and manages multiple turf pitch facilities.
- **TURF (1) ── (N) SLOT_HOLD:** A facility publishes discrete hourly slots; each hold record locks a specific time window.
- **USER (1) ── (N) BOOKING:** A customer initiates reservations that transition from pending advance payment to confirmed state.
- **BOOKING (1) ── (1) SLOT_HOLD:** Every active booking maintains a 1-to-1 association with a locked or confirmed slot.
- **BOOKING (1) ── (1) PAYMENT:** Records digital gateway transaction metadata, merchant TrxIDs, and IPN verification responses.
- **BOOKING (1) ── (1) REVIEW:** Allows a customer to submit a single verified rating once the match reaches `COMPLETED` status.

7. FUNCTIONAL REQUIREMENTS: MODULE 1

MODULE 1: AUTHENTICATION & IDENTITY MANAGEMENT

FR-AUTH-01: Multi-Role User Registration

• *Description:* Registers new accounts, assigning one of three roles: `CUSTOMER`, `TURF_OWNER`, or `ADMIN`. Platform Admins cannot be created via public APIs; they require seed scripts or direct database configuration.
• *Behavior:* Requires Full Name, valid Bangladesh phone number (+8801XXXXXXXXX), unique email, password, and target role. Passwords hashed using bcrypt (cost factor: 12).
• *Acceptance Criteria:* Rejects numbers not matching BD regex: `^(?:\+8801|01)[3-9]\d{8}$`; rejects passwords < 8 chars or lacking uppercase, lowercase, and numeric characters; returns HTTP 409 Conflict if email or phone is already registered.

FR-AUTH-02: Identity Authentication and Token Issuance

• *Description:* Authenticates credentials and issues signed JSON Web Tokens (JWT).
• *Behavior:* Uses a dual-token model: short-lived access token (15-min expiry) and cryptographically secure refresh token (7-day expiry) stored in an `httpOnly`, `SameSite=Strict`, secure cookie.
• *Acceptance Criteria:* Successful login returns user role and access token; sets refresh cookie; returns HTTP 401 Unauthorized for bad credentials without exposing email existence; rate limits to 5 failed attempts per 15 minutes per IP.

FR-AUTH-03: Role-Based Route Authorization

• *Description:* Middleware protects API endpoints based on authenticated role claims.
• *Behavior:* Inspects `Authorization: Bearer <token>`, decodes claims, verifies signature, and matches roles.
• *Acceptance Criteria:* Returns HTTP 401 if token is missing/expired; returns HTTP 403 Forbidden on role mismatch.

7. FUNCTIONAL REQUIREMENTS (CONT.)

MODULE 2: TURF PROFILE & DISCOVERY MANAGEMENT

FR-TURF-01: Turf Venue Registration by Owner

• *Description:* Turf owners submit venue profiles for administrative review.
• *Behavior:* Collects Turf Name, Description, Address, Dhaka Area (selected from predefined 25 zones), GeoJSON Point Coordinates, Available Pitch Formats (5, 6, 7-a-side), Amenities (Parking, Changing Rooms, Floodlights, Bibs, Boot Rentals, Cafeteria), and photos.
• *Acceptance Criteria:* Saves listing with `verificationStatus = PENDING`; requires latitude between 23.6500 and 23.9500 and longitude between 90.3000 and 90.5500 (Dhaka perimeter); max 5 images (JPG/PNG/WebP, max 5MB each).

FR-TURF-02: Public Turf Search and Geospatial Querying

• *Description:* Customers discover active turfs using text and geospatial queries.
• *Behavior:* Queries MongoDB via `$geoWithin` or `$nearSphere` against a `2dsphere` index, filtering by area, price, and amenities.
• *Acceptance Criteria:* Returns only turfs with `verificationStatus == APPROVED` and `isActive == true`; sub-200ms latency for collections up to 1,000 turfs; supports cursor/offset pagination (default: 12 per page).

MODULE 3: GEOSPATIAL DHAKA MAP EXPERIENCE

FR-MAP-01: Leaflet Vector Map Integration

• *Description:* Interactive Dhaka map displaying registered turfs, rendered via React Leaflet.
• *Behavior:* Centers on Dhaka (`[23.8103, 90.4125]`, zoom 12) with OpenStreetMap vector tiles. Selecting a marker displays a compact summary card (Name, Area, Price/Hr, Rating, and detail link) with Google Maps navigation link (`https://www.google.com/maps/dir/?api=1&destination=LAT,LNG`).
• *Acceptance Criteria:* Clusters overlapping markers via `react-leaflet-cluster`; syncs markers with current area/text filters; provides toggle between "Map View" and "List View" on screens < 768px.

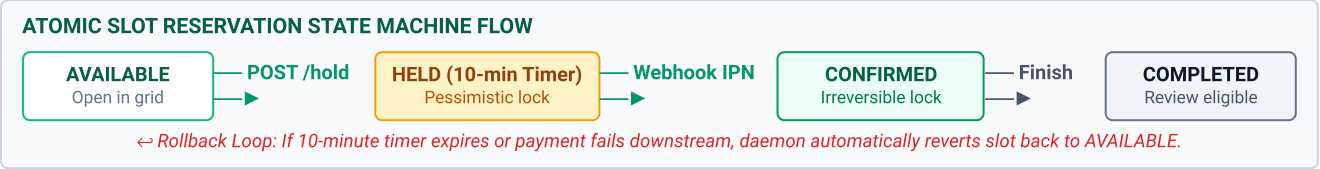
MODULE 4: SLOT ALLOCATION & CONCURRENCY CONTROL

FR-SLOT-01: Operating Schedule Configuration

• *Behavior:* Configures opening time, closing time, default slot length (60 mins), and pricing tiers (Daytime off-peak vs. Evening floodlit peak). Rejects schedules where closing precedes opening without explicit next-day offset flags (e.g., 6:00 PM to 2:00 AM). Automatically generates non-overlapping bookable time slots.

FR-SLOT-02: Atomic Slot-Hold Mechanism

• *Behavior:* Prevents double-bookings by placing a 10-minute hold on a slot during checkout. Backend creates a `SlotHold` document with `status = HELD` and `lockedUntil = now() + 10 mins` inside an atomic transaction with a unique compound index on `[turfId, date, startTime]`.
• *Acceptance Criteria:* Aborts and returns HTTP 409 Conflict if slot is held or booked: *"Slot is currently held by another user or already booked."* A background job executes every 60 seconds to revert expired holds to `AVAILABLE`.



7. FUNCTIONAL REQUIREMENTS (CONT.)

MODULE 5: CHECKOUT & ADVANCE PAYMENT ENGINE

FR-PAY-01: Payment Initiation and Transaction Tokenization

• *Description:* Initializes digital payments for required advances (20% to 50% of slot fee).

• *Behavior:* Creates gateway session (SSLCOMMERZ or bKash) using unique reference: `TRX-[NANOMARK]-[TIMESTAMP]`. Sandbox mock runner simulates flow for local grading.

• *Acceptance Criteria:* Validates that slot is actively `HELD` by user before redirecting; stores pending transaction record; rejects client requests attempting to alter advance amount.

FR-PAY-02: Server-Side Payment Verification (IPN)

• *Description:* Verifies payment status using secure server-side webhooks (Instant Payment Notification / IPN) rather than client-side redirects.

• *Behavior:* Validates incoming webhook HMAC signatures, triggers direct server-to-server gateway verification, and updates booking status.

• *Acceptance Criteria:* Client redirection never confirms booking without verified IPN; handles duplicate webhooks idempotently; transitions booking to `CONFIRMED` and slot to `BOOKED`; failed payments release slot immediately.

MODULE 6: REVIEWS & RATINGS

FR-REV-01 (Verified Reviews): Allows 1-to-5 star rating and 500-char review strictly for customers with a `COMPLETED` booking. Restricts each booking to exactly one review; recalculates turf average atomically via aggregation pipeline. Allows owners to flag reviews for moderation.

MODULE 7: COMPLIANCE & ADMIN

FR-ADM-01 (Turf Moderation): Displays pending venues queue with photos, trade licenses, coordinates, and contact details. Admins can `APPROVE` or `REJECT` (with mandatory feedback note), or `SUSPEND` active listings immediately if complaints arise.

8. DATA MODEL AND SCHEMA SPECIFICATIONS

Document database running on MongoDB Atlas with replica sets, schema validations, and compound indexes.

8.1 User Schema (`users`)

```
{ "_id": "ObjectId", "fullName": "String (required, trim, max 80)", "email": "String (required, unique, lowercase, index)", "phone": "String (required, unique, index, matches BD regex)", "passwordHash": "String (required, bcrypt cost: 12)", "role": "String (enum: ['CUSTOMER', 'TURF_OWNER', 'ADMIN'], default: 'CUSTOMER')", "isSuspended": "Boolean (default: false)", "createdAt": "ISODate", "updatedAt": "ISODate" } // Indexes: { email: 1 } (unique), { phone: 1 } (unique)
```

8.2 Turf Schema (`turfs`)

```
{ "_id": "ObjectId", "ownerId": "ObjectId (ref: 'users', required, index)", "name": "String (required, trim, max 100)", "description": "String (required, max 1500)", "address": "String (required)", "area": "String (required, enum: 25 Dhaka Zones)", "location": { "type": "String (enum: ['Point'], default: 'Point')", "coordinates": ["Number (Lng)", "Number (Lat)"] }, "dimensions": "String (e.g., '7-a-side: 40m x 20m')", "amenities": ["String (PARKING, CHANGING_ROOM, FLOODLIGHTS, ...)"], "images": [{ "url": "String", "publicId": "String", "isHero": "Boolean" }], "hourlyRate": "Number (required, min 500, max 15000)", "advancePercentage": "Number (required, min 10, max 100, default: 30)", "verificationStatus": "String (enum: ['PENDING', 'APPROVED', 'REJECTED'], default: 'PENDING')", "ratingSummary": { "average": "Number (0-5)", "reviewCount": "Number" }, "isActive": "Boolean (default: true)", "createdAt": "ISODate", "updatedAt": "ISODate" } // Indexes: { location: "2dsphere" }, { area: 1, verificationStatus: 1, isActive: 1 }
```

8. DATA MODEL AND SCHEMA SPECIFICATIONS (CONT.)

8.3 Slot Hold Schema (slot_holds)

```
{ "_id": "ObjectId", "turfId": "ObjectId (ref: 'turfs', required)", "slotDate": "String ('YYYY-MM-DD')", "startTime": "String ('HH:mm')", "endTime": "String ('HH:mm')", "status": "Enum ['AVAILABLE', 'HELD', 'BOOKED', 'BLOCKED']", "heldByUserId": "ObjectId (ref: 'users')", "lockedUntil": "ISODate (TTL)", "createdAt": "ISODate" } // Compound Unique Index: // { turfId: 1, slotDate: 1, startTime: 1 } // Lookup: { status: 1, lockedUntil: 1 }
```

8.4 Booking Schema (bookings)

```
{ "_id": "ObjectId", "bookingRef": "String (UQ, 'BK-YYYYMMDD-XXXX')", "customerId": "ObjectId (ref: 'users')", "turfId": "ObjectId (ref: 'turfs')", "slotDate": "String ('YYYY-MM-DD')", "startTime": "String ('HH:mm')", "totalPrice": "Number", "advanceAmount": "Number", "dueAmountAtVenue": "Number", "status": "Enum ['PENDING_PAYMENT', 'CONFIRMED', 'CANCELLED', 'COMPLETED']", "paymentId": "ObjectId (ref: 'payments')" } // Indexes: { customerId: 1, status: 1 }, // { bookingRef: 1 } (unique)
```

8.5 Payment Ledger (payments)

```
{ "_id": "ObjectId", "bookingId": "ObjectId (ref: 'bookings')", "gatewayTransactionId": "String (UQ)", "paymentGateway": "Enum ['SSLCOMMERZ', 'BKASH', 'MOCK_SANDBOX']", "amountPaid": "Number", "currency": "String (default: 'BDT')", "gatewayResponsePayload": "Object (raw)", "paymentStatus": "Enum ['INITIATED', 'SUCCESS', 'FAILED']", "verifiedAt": "ISODate" } // Indexes: { gatewayTransactionId: 1 } (UQ)
```

9. RESTFUL API ENDPOINTS: CORE MODULES

Standard response envelope: `{"success": true, "data": {}, "message": "...", "errors": null}`

9.1 Authentication Endpoints (/api/v1/auth)

Method / Path	Description	Auth Level	Important Payload Params	Status
POST /register	Register new account	Public	fullName, email, phone, password, role	201, 400, 409
POST /login	Authenticate user & get token	Public	email, password	200, 401, 429
POST /refresh-token	Renew expired access token	Refresh Cookie	(Cookie-based session)	200, 401
POST /logout	Invalidate active session	Bearer Token	(None)	200, 401

9.2 Turf Discovery & Query Endpoints (/api/v1/turfs)

Method / Path	Description	Auth Level	Query / Body Parameters	Status
GET /	Paginated search & filter	Public	?area=Mirpur&minPrice=1000&page=1	200, 400
GET /map-pins	Map-ready coordinates list	Public	?swLat=23.7&swLng=90.3&neLat=23.9	200, 400
GET /:id	Detailed turf profile	Public	params: { id }	200, 404
POST /	Register new turf facility	TURF_OWNER	name, address, area, coords, price	201, 400, 403
PATCH /:id	Update owned turf details	TURF_OWNER	name, description, amenities, rate	200, 403, 404

9.3 Slot & Availability Engine (/api/v1/slots)

9. RESTFUL API ENDPOINTS (CONT.)

9.4 Bookings & Payment Workflows (`/api/v1/bookings`)

Method / Path	Description	Auth Level	Query / Body Parameters	Status
`POST` /checkout	Initialize payment flow	CUSTOMER	{ slotHoldId, gatewayChoice }	200, 400, 410
`POST` /payment-webhook	Gateway IPN confirmation	Public (HMAC)	Gateway specific callback payload	200, 400, 403
`GET` /my-bookings	Retrieve user booking history	CUSTOMER	?page=1&status=CONFIRMED	200, 401
`GET` /owner/overview	Turf owner reservations grid	TURF_OWNER	?turfId=...&date=YYYY-MM-DD	200, 403

9.5 Administration Endpoints (`/api/v1/admin`)

Method / Path	Description	Auth Level	Query / Body Parameters	Status
`GET` /pending-turfs	View venues awaiting approval	ADMIN	?page=1	200, 403
`PATCH` /verify-turf	Approve or reject a listing	ADMIN	{ turfId, status, rejectionReason }	200, 400, 403
`PATCH` /toggle-ban	Suspend user or turf listing	ADMIN	{ targetEntity, entityId, state }	200, 403, 404

10. UI/UX SPECIFICATIONS AND INFORMATION ARCHITECTURE

10.1 DESIGN SYSTEM

- Slate Dark: #0F172A (Headers)
- Turf Emerald: #10B981 (Brand)
- Peak Amber: #F59E0B (Peak slots)
- Error Red: #EF4444 (Held/Expired)

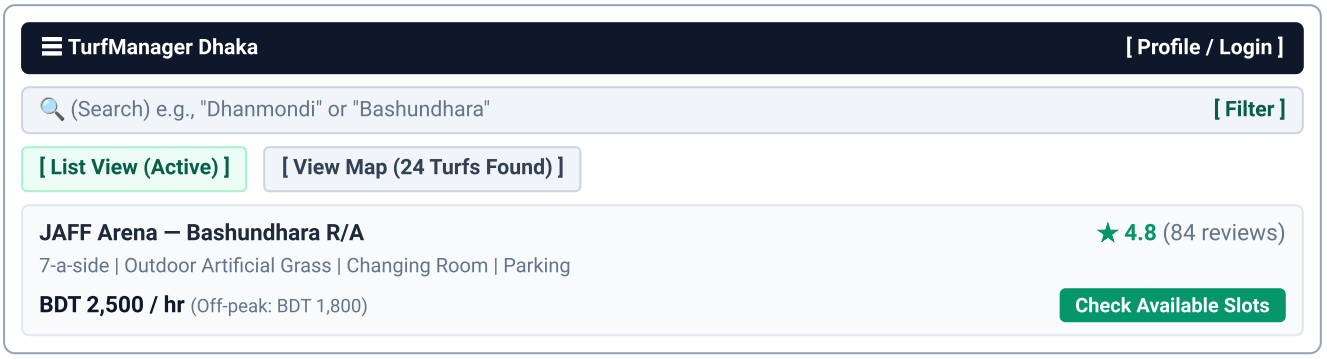
TYPOGRAPHY & A11Y

- Inter / Plus Jakarta Sans
- Base font: 16px (no iOS auto-zoom)
- Contrast: WCAG AA (≥ 4.5:1)
- Visible interactive focus states

ROLE SITEMAP PATHS

- *Customer:* /turfs, /my-bookings
- *Owner:* /owner/turfs/:id/slots
- *Admin:* /admin/moderation
- *Public:* Home, Discovery, Map

10.2 Page-by-Page Specifications & Mobile Discovery Wireframe



Page SPEC-01 (Turf Discovery & Dhaka Map): Path: `/turfs` • Public access • Sticky filter bar, list/map toggle, marker clusters, bottom preview drawer on mobile linking to Google Maps directions.

Page SPEC-02: Date Selection & Slot Booking Flow (`/turfs/:id/book`)
Authenticated Customer access • Horizontal 7-day date selector • Interactive time slot grid separated into Daytime (6:00 AM – 5:00 PM) and Evening Floodlit (5:00 PM – 12:00 AM) • Sticky bottom checkout drawer.



Page SPEC-03: Booking Checkout & Payment
Path: `/checkout/:holdId` • Customer • 10-minute countdown timer (mm:ss), breakdown (Base, Advance Due, Venue Remaining), gateway selector (bKash, SSLCOMMERZ, Mock Sandbox). Reaching 00:00 triggers auto-release and expiry modal.

Page SPEC-04: Owner Schedule Management
Path: `/owner/turfs/:id/slots` • Turf Owner • Calendar matrix view, bulk slot generator, manual block/unblock toggles for maintenance or offline phone walk-ins to keep schedules aligned.

11. NON-FUNCTIONAL REQUIREMENTS (NFR)

Metric Category	Production Target Specification	Enforcement Mechanism
Performance (P95)	Static assets < 1.5s on 4G. Dynamic endpoints < 250ms under 50 CCU.	2dsphere indexes, CDN WebP images
System Availability	99.0% operational uptime throughout the academic term.	Express error-boundaries, PM2
Accessibility	Conforms to WCAG 2.1 Level AA (4.5:1 contrast, 44×44px touch targets).	Tailwind touch sizing, semantic HTML
Security Standards	bcrypt (12 rounds), JWT httpOnly cookies, Helmet headers, TLS 1.3.	Express middleware, env secrets
Data Concurrency	Zero double-bookings under parallel race load.	MongoDB ACID sessions & unique indexes

12. BUSINESS RULES MATRIX

ID	Rule Description	Enforcement Layer	Failure Action
BR-01	Only approved and active venues appear in public searches.	Database Query / API	Hidden from customer search results.
BR-02	Turf owners can only edit their own venues.	Middleware (verifyOwnership)	Returns 403 Forbidden + audit log.
BR-03	Slots cannot be booked in the past.	Validation Schema	Rejects requests before Date.now().
BR-04	Two users cannot check out the same slot simultaneously.	Database Unique Index	Second request returns 409 Conflict.
BR-05	Checkout hold expires after exactly 10 minutes.	Scheduled TTL / Daemon	Slot status reverts to AVAILABLE.
BR-06	Payment requires server-side IPN verification.	Payment Controller	Redirection alone does not confirm booking.
BR-07	Reviews require a completed booking.	Controller Query	Rejects review if no COMPLETED record.
BR-08	Users can only review a booking once.	Database Unique Index	Second review attempt

12.2 EDGE CASES AND SYSTEM RECOVERY

- **User drops network during redirect:** Slot remains HELD until 10-minute timer expires. If payment succeeds downstream, webhook IPN updates booking to CONFIRMED. If failed, cleanup returns slot to AVAILABLE.
- **Webhook arrives before user returns:** Webhook updates DB state atomically; when user reconnects, frontend displays confirmation.
- **Manager suspends pitch with active bookings:** Existing CONFIRMED bookings remain valid; new holds/bookings blocked immediately.

13. IMPLEMENTATION PHASES & 14. TESTING STRATEGY

DEVELOPMENT ROADMAP

- **Phase 1: Core MVP (Wks 1–4):** Multi-role Auth, Turf Profiles, Leaflet Map, 10-Min Pessimistic Lock, Mock Sandbox Gateway.
- **Phase 2: Maturation (Wks 5–8):** Live SSLCOMMERZ/bKash IPN, Verified Reviews, Owner Overrides, Expiration Cron.
- **Phase 3: Future Work:** In-App Split-Billing, PWA Offline Access, Multi-Sport Facility Expansion.

TESTING PYRAMID & CORE CASES

- **TC-CONCUR-01 (Parallel Hold):** Concurrent identical hold requests via Promise.all. Exact 1 succeeds (200), other receives 409 Conflict.
- **TC-PAY-01 (IPN Webhook):** Call webhook with invalid HMAC: returns 403 Forbidden; valid signature confirms booking.
- **TC-AUTH-01 (Role Escalation):** Customer calling PATCH /admin returns 403 Forbidden.

15. ACADEMIC PROJECT PLANNING & RESPONSIBILITIES

Work Breakdown Structure (WBS):

- **Sprint 1:** Project repo setup, Tailwind, MongoDB Atlas, Auth/JWT.
- **Sprint 2:** Turf models, Cloudinary uploads, Leaflet map view.
- **Sprint 3:** Schedule generator, atomic slot-hold transaction, TTL jobs.
- **Sprint 4:** Mock payment engine, SSLCOMMERZ IPN, confirmations.
- **Sprint 5:** Owner overrides, admin moderation, verified reviews.
- **Sprint 6:** Edge tests, mobile polish, Swagger docs, capstone presentation.

Team Roles & Responsibilities:

- **Full-Stack Lead:** Architecture, DB schema, slot concurrency, payment webhooks, deployment.
- **Frontend & UI/UX:** Tailwind layouts, Leaflet map view, slot selector, checkout flow.
- **Backend & APIs:** Auth/RBAC, search queries, turf profiles, admin moderation endpoints.
- **QA & Docs:** Unit/integration tests, Postman collection, academic report, presentation demo.

16. TECHNICAL RISK REGISTER

Risk Event	Severity	Probability	Mitigation Strategy
Merchant Account Delays	High	High	Deliver Mock Gateway runner matching interface for grading and demos.
Race Conditions During Booking	Critical	Medium	Use MongoDB multi-document transactions and unique compound indexes.
Map Rendering Performance	Medium	Medium	Use marker clustering and limit queries to visible map bounding box.
Broken Third-Party Media Links	Medium	Low	Store images in Cloudinary with fallback placeholder graphics.

17. EVALUATION CRITERIA & 18. ASSUMPTIONS

ACADEMIC EVALUATION CRITERIA

[+] **Architecture:** Clean MVC/Service separation. [+] **Concurrency:** Atomic database holds preventing simultaneous bookings. [+] **UX:** Mobile-first Dhaka map with responsive touch targets. [+] **Security:** Hashed passwords, protected cookies, role checks.

OPEN DECISIONS & NEXT STEPS

Assumptions: (1) Mock Sandbox gateway for grading, (2) Fixed 60-min slots starting on the hour, (3) OpenStreetMap default tiles.
Next Steps: (1) MongoDB Atlas initialization, (2) Express API scaffolding, (3) Vite/Tailwind frontend shell.